UnB | PPCA
Professional Master's in Applied Computing

Digital Twins for Recycling Cooperatives

An Integrated Multi-Agent Approach for Traceability, Efficiency and Sustainability

P2B – Reconcilication and Refinement of the R3 Groups' Proposals

Group R3 — P2B

R3S1 – Traceability and Transparency
R3S2 – Operational Efficiency
R3S3 – Sustainability and Social Impact

1 Context and Motivation

Recycling cooperatives play an essential role in the circular economy, but face challenges in traceability, operational efficiency and social recognition.



- Low visibility
- Manual processes
- Limited traceability
- Insufficient social recognition

2 Integrated System View



3 Reconciliation of Proposals (R3)

Integration and refinement of the contributions from the three groups, without hierarchy, in a single and complementary model.

R3S1 Traceability and Transparency
- Material flow registration
- Provenance and certifications
- Transparency for stakeholders

R3S2 Operational Efficiency
- Process optimization
- Resource and capacity management
- Cost and loss reduction

R3S3 Sustainability and Social Impact
- Environmental indicators
- Social impact and inclusion
- Valuing the work of cooperatives

Integrated Model (P2B)
- Systemic and complementary vision
- Consolidated functional and non-functional requirements
- Basis for the multi-agent architecture

4 Identified Capabilities

System capabilities organized according to **WHAT** (what) and **HOW** (how), resulting from the integration of the three groups.

WHAT (system capabilities)	HOW (technological enablers)
Material traceability	Digital twins and simulation
Operational monitoring	Specialized AI agents
Sustainability indicators	Data integration
Decision support for cooperatives	Interfaces and visualizations
Stakeholder engagement	Governance and security

5 Multi-Agent Architecture

System with four collaborative agents, each with specific responsibilities and interactions to achieve the system's objectives.



6 Number of Agents

Comparative analysis and justification of the choice of four agents.

3 agents
- Simpler architecture
- Lower complexity
- Risk of role overlap
- Limited separation of responsibilities

4 agents (adopted option)
- Covers all identified capabilities
- Balance between complexity and benefits
- Scalable and modular
- Better separation of responsibilities

5+ agents
- Higher granularity
- Greater complexity
- Higher engineering overhead at this stage

7 Real–Digital Convergence

Iterative process of evaluation and continuous improvement between the digital system and the reality of cooperatives.



8 Expected Contributions and Results

Integrated contributions from P2B aligned with the Sustainable Development Goals (SDGs).

Technological
- Multi-agent architecture for digital twins
- Integration of heterogeneeus data
- Decision support based on simulation and AI

Social
- Strengthening cooperatives
- Inclusion and valorization of waste pickers
- Greater transparency and social recognition

Environmental
- Increased recycling
- Reduced environmental impacts
- Contribution to the circular economy

Related SDGs: 8 DECENT WORK AND ECONOMIC GROWTH · 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE · 11 SUSTAINABLE CITIES AND COMMUNITIES · 12 RESPONSIBLE CONSUMPTION AND PRODUCTION · 13 CLIMATE ACTION

9 Next Steps

1. Detailed architecture design
2. Validation with cooperatives
3. Prototype development
4. Evaluation of results
5. Scaling and replication

Technology, people and environment hand in hand for a more inclusive and circular economy.